

Quiz Answer

Linear Regression - Gradient Descent (Second Iteration)

From the previous iteration:

$$w = 0.182 \quad b = 0.0645 \quad \text{learning rate} = 0.01$$

Data

x	1	2	3	4
y	2	2.8	3.6	4.5

1. Calculate Predictions

$$y_{\text{hat}} = w \cdot x + b$$

$$x = 1: y_{\text{hat}} = 0.182(1) + 0.0645 = 0.2465$$

$$x = 2: y_{\text{hat}} = 0.182(2) + 0.0645 = 0.4285$$

$$x = 3: y_{\text{hat}} = 0.182(3) + 0.0645 = 0.6105$$

$$x = 4: y_{\text{hat}} = 0.182(4) + 0.0645 = 0.7925$$

2. Gradient with respect to w

$$dL/dw = (2/4) * \text{sum}((y_{\text{hat}} - y) * x)$$

$$dL/dw = -15.1475$$

3. Gradient with respect to b

$$dL/db = (2/4) * \text{sum}(y_{\text{hat}} - y)$$

$$dL/db = -5.411$$

4. Update Parameters

$$w_{\text{new}} = w - \text{alpha} * dL/dw$$

$$w_{\text{new}} = 0.182 - 0.01(-15.1475) = 0.333475$$

$$b_{\text{new}} = b - \text{alpha} * dL/db$$

$$b_{\text{new}} = 0.0645 - 0.01(-5.411) = 0.11861$$

Final Answer

New w	0.333475
New b	0.11861